Brass Nut Pendulum Swing

Run r_20261101_044831_cf0467 · Date 2026-11-01 · Verdict: PASS WITH WARNINGS

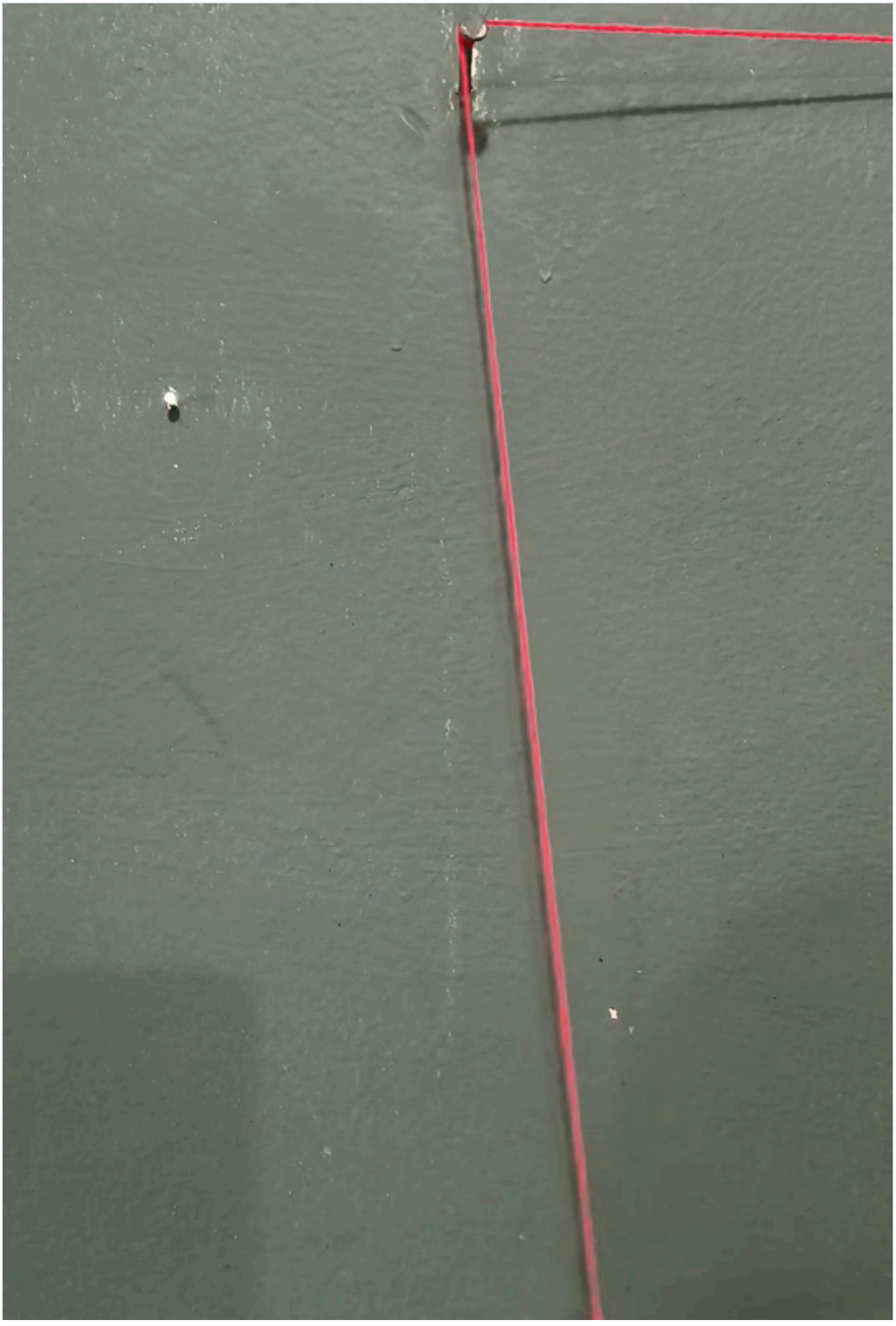
$g = 9.65 \pm 0.12\ m/s^2$

Aim

The aim was to measure how fast a hanging brass nut swings and to use that timing to estimate gravity, and to test the predicted period.

Setup

A small brass nut hangs from a red string pinned to a green wall and swings as a pendulum.





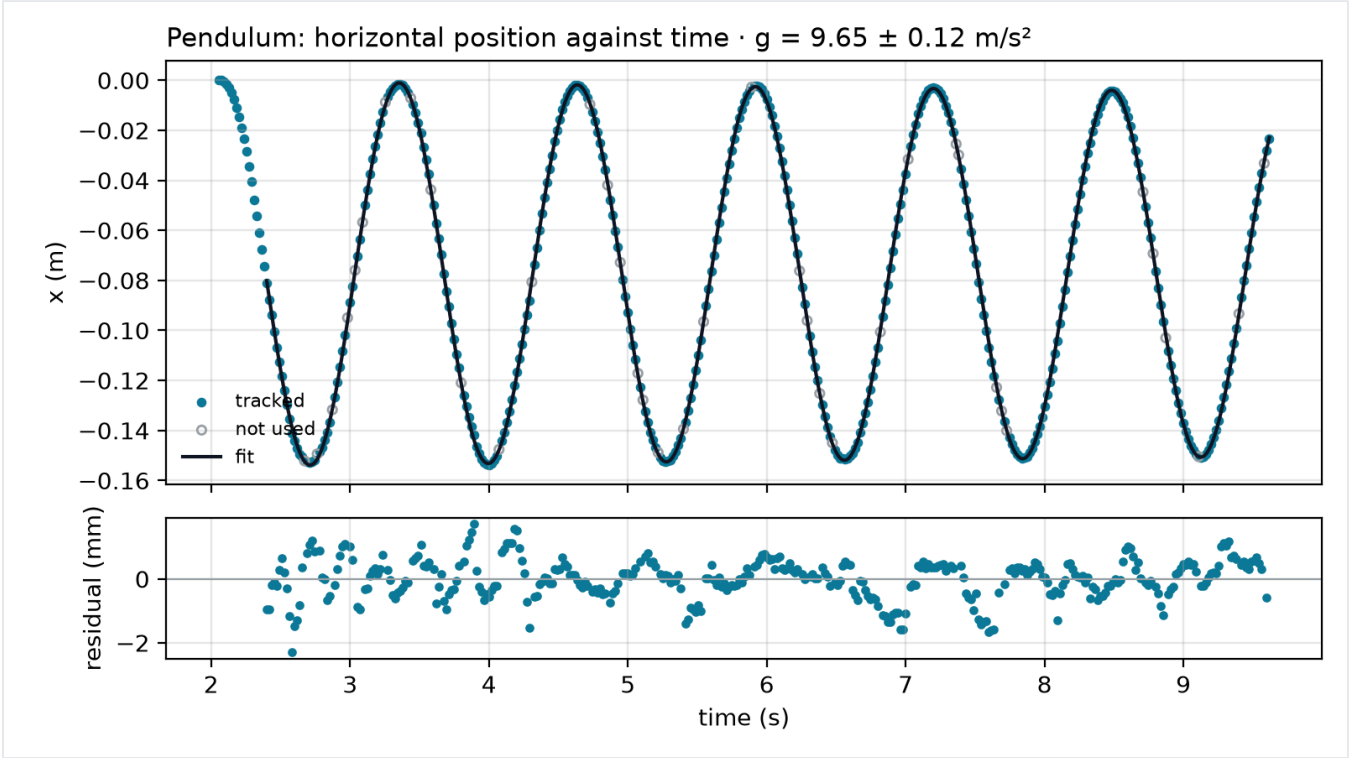
Key frame

Method

Tracking gives the nut position in each video frame. An automatic window trims hand contact and keeps steady swinging. A damped sine curve is fitted to the sideways motion, which captures the repeating swing that slowly shrinks. Repeated resampling gives the confidence range for timing. String length turns the pixel swing into physical length, and the pendulum relation then gives gravity from length and period.

Model	Role in this run
Muse Voice Transcribe	speech → transcript
Muse Spark 1.3	Planner, Analyst, Auditor, Translator
SAM 3.1	object tracking on the video and the ruler
Muse Wonder 1.0	not used in this run
Muse Spark Open (open weights, via Baseten)	Designer

Results



Trajectory

Swing timing

The swing repeats regularly. The measured period is period = 1.28236 ± 0.00017 s and the amplitude corrected period is period0 = 1.2795 ± 0.0014 s. The bootstrap check agrees period = 1.2795 ± 0.0014 s.

Gravity estimate

From timing and string length, gravity is g = 9.65 ± 0.12 m/s². Comparison with the reference gives a difference of -1.39% from the reference value.

Fit and scale

The model follows the trajectory very closely The damped sine model fits the swing trajectory very closely with small residuals.. The scale rests on spoken length and pivot geometry scale = 0.485 mm/px.

Hypothesis evaluation

The predicted period is longer than the measured swing period = 1.28236 ± 0.00017 s, and the formal test does not support the prediction not supported.

Error analysis

Main limits come from the length scale, which rests only on the spoken string length and pivot geometry scale = 0.485 mm/px, so any error in stated length moves gravity directly. The swing is fairly wide, so air drag and amplitude effects matter, handled by the corrected timing period0 = 1.2795 ± 0.0014 s. Tracking noise is small and the model follows the motion closely The damped sine model fits the swing trajectory very closely with small residuals., but residuals show a smooth pattern, meaning the simple damped sine is not perfect. The bootstrap range period = 1.2795 ± 0.0014 s allows for that pattern and gives an honest spread.

Conclusion

The swing is faster than predicted, so the prediction is not supported not supported. The gravity estimate is close to the reference a difference of -1.39% from the reference value with value g = 9.65 ± 0.12 m/s². Within the stated scale and fitting limits, the pendulum timing gives a consistent result.

Next experiment

Period vs String Length: Testing T = 2π√(L/g) at Home / Periode vs Panjang Tali: Menguji T = 2π√(L/g) di Rumah

The predicted period for the shortest tested length is 1.43 s, for the second tested length is 2.02 s, for the third tested length is 2.48 s, and for the longest tested length is 2.86 s, consistent with period increasing as the square root of length.

- 0.500 m → 1.43 s
- 1.00 m → 2.02 s
- 1.50 m → 2.48 s
- 2.00 m → 2.86 s

Controlled variables: release angle (small, ~5-10° same for all trials) / sudut pelepasan (kecil, ~5-10° sama untuk semua percobaan), bob mass (same bob) / massa beban (beban yang sama), measurement method and timer / metode pengukuran dan pewaktu

1. Tie a small heavy bob (e.g., nut, bolt, or small toy) to a thin string. Measure length from pivot edge to centre of bob with a tape measure.
2. Set up pivot: tape string firmly to top of door frame or sturdy table edge, allowing free swing without hitting walls. Keep total length ≤2.0 m, heights ≤2 m. Ask an adult to help secure it.
3. Set length 1: Adjust string to 0.5 m total length. Mark string at pivot point with pen/tape so you can reset.

4. Pull bob sideways only a small angle (~5-10°, about a hand width sideways for 1 m string). Release gently without pushing. Let it swing.
5. Measure period: Use phone stopwatch. Time 10 complete back-and-forth swings (start/stop on same side). Divide total time by 10 to get period T. Record.
6. Repeat step 5 three times for 0.5 m and take average.
7. Change ONLY length: Lengthen string to 1.0 m (re-measure pivot to bob centre), keep same bob, same small release angle. Repeat timing (3 x 10 swings, average).
8. Repeat for 1.5 m and then 2.0 m, using identical procedure. Do not change bob or swing angle.
9. Compare measured periods to predictions and plot T^2 vs L to check linear relationship.

Safety: Household objects only; no fire, no mains electricity, no glass. · Keep heights at most 2 m; use a stable pivot (door frame), do not climb on unstable chairs; adult supervision required. · Children must be supervised by an adult for the entire experiment. · Use small, light bob; ensure area below is clear so bob cannot hit people or break objects. · Swing with small angle only to avoid large forces and hitting nearby objects.

Appendix A: Claim Ledger

#	Claim	Evidence
c01	The measured gravitational acceleration is 9.65 ± 0.12 m/s² with confidence interval [9.41, 9.89] m/s².	der_04
c02	The measured oscillation period is 1.28236 ± 0.00017 s with confidence interval [1.28202, 1.28269] s.	der_01
c03	The amplitude corrected period is 1.2795 ± 0.0014 s with confidence interval [1.2767, 1.2823] s.	der_02
c04	The measured gravity differs from the reference by -1.39% with measured value 9.65 ± 0.12 m/s².	cmp_01
c05	The hypothesis evaluation outcome is not supported.	cmp_02
c06	The damped sine model fits the swing trajectory very closely with small residuals.	fit_01
c07	The bootstrap estimate of the period is 1.2795 ± 0.0014 s with confidence interval [1.2767, 1.2823] s.	ci_01
c08	The length scale comes from the spoken string length and the pivot geometry.	cal_02

Appendix B: Gate table

Gate	Status	Diagnosis	Message
G1 Track integrity hard	pass		Passed.
G2 Scale consistency soft	warn	single_scale	Only one scale method is available, so scale consistency cannot be checked.
G3 Fit quality soft	warn	smooth_model_mismatch	The fit is very close (R^2 0.9999, RMS 1.29 px), but the residuals still follow a smooth pattern (Durbin–Watson 0.31), as a wide swing or air drag causes. The uncertainty is computed with a block bootstrap, which allows for this.
G4 Physical plausibility soft	pass		Passed.
G5 Ledger completeness hard	pass		Passed.

Appendix C: Data sample

t (s)	x (m)	y (m)	valid	flag
2.054545	0.000000	0.000000	1	ok
2.345455	-0.060784	0.003946	1	ok
2.636364	-0.149369	-0.002735	1	ok
2.927273	-0.114553	0.002465	1	ok
3.218182	-0.016970	0.003590	1	ok
3.509091	-0.021007	0.002185	1	ok
3.800000	-0.120946	0.001534	0	outlier
4.090909	-0.144122	-0.001990	1	ok
4.381818	-0.053367	0.004898	1	ok
4.672727	-0.003284	-0.001809	1	ok
4.963636	-0.079623	0.003478	1	ok

t (s)	x (m)	y (m)	valid	flag
5.254545	-0.152052	-0.003896	1	ok
5.545455	-0.096521	0.004416	0	outlier
5.836364	-0.008464	0.002040	1	ok
6.127273	-0.038106	0.003251	1	ok
6.418182	-0.134519	-0.001864	1	ok
6.709091	-0.133207	0.001057	1	ok
7.000000	-0.037497	0.005849	1	ok
7.290909	-0.010125	0.000117	1	ok
7.581818	-0.100229	0.002872	1	ok
7.872727	-0.150322	-0.003327	1	ok
8.163636	-0.076775	0.004441	1	ok
8.454545	-0.004942	-0.001327	1	ok
8.745455	-0.056890	0.002858	1	ok
9.036364	-0.143404	-0.004721	1	ok
9.327273	-0.116281	0.002958	1	ok
9.618182	-0.023470	0.001793	1	ok

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